

Foundation Mathematics 1017SCG
Week 3 Workshop Solutions

Remember that the method shown in the solutions is not the only way to solve the question. There are usually multiple ways to solve the same question.

1. (a) $30 \times \frac{\pi}{180} = \frac{\pi}{6}$
- (b) $45 \times \frac{\pi}{180} = \frac{\pi}{4}$
- (c) $60 \times \frac{\pi}{180} = \frac{\pi}{3}$
- (d) $90 \times \frac{\pi}{180} = \frac{\pi}{2}$
- (e) $210 \times \frac{\pi}{180} = \frac{7\pi}{6}$
- (f) $300 \times \frac{\pi}{180} = \frac{5\pi}{3}$

2. (a) $\frac{2\pi}{3} \times \frac{180}{\pi} = 120^\circ$
- (b) $\frac{5\pi}{6} \times \frac{180}{\pi} = 150^\circ$
- (c) $\frac{5\pi}{4} \times \frac{180}{\pi} = 225^\circ$
- (d) $\frac{3\pi}{2} \times \frac{180}{\pi} = 270^\circ$
- (e) $\pi \times \frac{180}{\pi} = 180^\circ$
- (f) $2 \times \frac{180}{\pi} = 114.59^\circ$

3. It is difficult to explain the working for this question as it involves the special triangles and the unit circle. If you have questions about the solutions, please ask during a tutorial.

θ	$\sin(\theta)$	$\cos(\theta)$	$\tan(\theta)$
30°	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{3}}$
45°	$\frac{1}{\sqrt{2}}$	$\frac{1}{\sqrt{2}}$	1
120°	$\sin(120^\circ) = \sin(60^\circ)$ $\frac{\sqrt{3}}{2}$	$\cos(120^\circ) = -\cos(60^\circ)$ $-\frac{1}{2}$	$\tan(120^\circ) = -\tan(60^\circ)$ $-\sqrt{3}$
240°	$\sin(240^\circ) = -\sin(60^\circ)$ $-\frac{\sqrt{3}}{2}$	$\cos(240^\circ) = -\cos(60^\circ)$ $-\frac{1}{2}$	$\tan(240^\circ) = \tan(60^\circ)$ $\sqrt{3}$
315°	$\sin(315^\circ) = -\sin(45^\circ)$ $-\frac{1}{\sqrt{2}}$	$\cos(315^\circ) = \cos(45^\circ)$ $\frac{1}{\sqrt{2}}$	$\tan(315^\circ) = -\tan(45^\circ)$ -1

4.

θ	$\sin(\theta)$	$\cos(\theta)$	$\tan(\theta)$
$\frac{\pi}{3}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\sqrt{3}$
$\frac{3\pi}{4}$	$\sin(\frac{3\pi}{4}) = \sin(\frac{\pi}{4})$ $\frac{1}{\sqrt{2}}$	$\cos(\frac{3\pi}{4}) = -\cos(\frac{\pi}{4})$ $-\frac{1}{\sqrt{2}}$	$\tan(\frac{3\pi}{4}) = -\tan(\frac{\pi}{4})$ -1
$\frac{4\pi}{3}$	$\sin(\frac{4\pi}{3}) = -\sin(\frac{\pi}{3})$ $-\frac{\sqrt{3}}{2}$	$\cos(\frac{4\pi}{3}) = -\cos(\frac{\pi}{3})$ $-\frac{1}{2}$	$\tan(\frac{4\pi}{3}) = \tan(\frac{\pi}{3})$ $\sqrt{3}$
$\frac{11\pi}{6}$	$\sin(\frac{11\pi}{6}) = -\sin(\frac{\pi}{6})$ $-\frac{1}{2}$	$\cos(\frac{11\pi}{6}) = \cos(\frac{\pi}{6})$ $\frac{\sqrt{3}}{2}$	$\tan(\frac{11\pi}{6}) = -\tan(\frac{\pi}{6})$ $-\frac{1}{\sqrt{3}}$

5.

θ	$\sin(\theta)$	$\cos(\theta)$	$\tan(\theta)$
90°	1	0	undefined
π	0	-1	0
2π	0	1	0
$-\frac{\pi}{4}$	$\sin(-\frac{\pi}{4}) = -\sin(\frac{\pi}{4})$ $-\frac{1}{\sqrt{2}}$	$\cos(-\frac{\pi}{4}) = \cos(\frac{\pi}{4})$ $\frac{1}{\sqrt{2}}$	$\tan(-\frac{\pi}{4}) = -\tan(\frac{\pi}{4})$ -1
$-\frac{5\pi}{6}$	$\sin(-\frac{5\pi}{6}) = -\sin(\frac{\pi}{6})$ $-\frac{1}{2}$	$\cos(-\frac{5\pi}{6}) = -\cos(\frac{\pi}{6})$ $-\frac{\sqrt{3}}{2}$	$\tan(-\frac{5\pi}{6}) = \tan(\frac{\pi}{6})$ $\frac{1}{\sqrt{3}}$

6. (a) $3x + 5x = 8x$
 (b) $9x^2 - 2x^2 = 7x^2$
 (c) $2x + 4x + 5y = 6x + 5y$ (Note that $6x$ and $5y$ are **unlike terms**)
 (d) $6x^2 - 5x^2 + 9x = x^2 + 9x$ (Note that x^2 and $9x$ are **unlike terms**)
 (e) The expression $5x^2 - 4x + 8$ cannot be simplified further
 (f) $9x - 5y + 3x + 4y = 9x + 3x - 5y + 4y = 12x - y$

7.

$$10a^3 - 6a^2 + 4a + 13$$

- (a) There are 4 terms in the expression.
 (b) The constant term is 13.
 (c) The coefficient of a^2 is -6 .
 (d) Given that $a = 2$:

$$\begin{aligned}
 10 \times 2^3 - 6 \times 2^2 + 4 \times 2 + 13 &= 10 \times 8 - 6 \times 4 + 4 \times 2 + 13 \\
 &= 80 - 24 + 8 + 13 \\
 &= 77
 \end{aligned}$$

(e) Given that $a = -1$

$$\begin{aligned} 10 \times (-1)^3 - 6 \times (-1)^2 + 4 \times (-1) + 13 &= 10 \times -1 - 6 \times 1 + 4 \times -1 + 13 \\ &= -10 - 6 - 4 + 13 \\ &= -7 \end{aligned}$$

Solutions